US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250280946
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	PARK; Minhoo et al.

ADAPTER ACCESSORY AND ELECTRONIC DEVICE CASE INCLUDING SAME

Abstract

example embodimentAn adapter accessory may include an accessory body including a coupling portion configured to couple an external accessory thereto, a deformable portion extending from the accessory body and configured to be at least partially deformable by a force applied from an outside, a button portion connected to the deformable portion and configured to press the deformable portion by the force applied from the outside or to be pressed by the deformable portion, and a fixing portion configured to be coupled to an external case and formed to protrude from at least a portion of the button portion. A material of the deformable portion may be the same as a material of the button portion. Various other embodiments may also be possible.

Inventors: **PARK; Minhoo (Suwon-si, KR), IM; Junghyuck (Suwon-si, KR), HA; Dohyung (Suwon-si, KR), KWON; Kijin (Suwon-si, KR), KIM; Kisung (Suwon-si, KR), PARK; Sangil (Suwon-si, KR), LEE; Wooseok (Suwon-si, KR), JUNG; Gwangchae (Suwon-si, KR), CHUNG; Chongyoon (Suwon-si, KR)**

Applicant: **Samsung Electronics Co., Ltd.** (Suwon-si, KR)

Family ID: **91324311**

Appl. No.: **19/217271**

Filed: **May 23, 2025**

Foreign Application Priority Data

KR	10-2022-0162476	Nov. 29, 2022
KR	10-2022-0171924	Dec. 09, 2022

Related U.S. Application Data

parent WO continuation PCT/KR2023/016409 20231020 PENDING child US 19217271

Publication Classification

Int. Cl.: **A45F5/00** (20060101); **A45F5/10** (20060101); **G06F1/16** (20060101)

U.S. Cl.:

CPC **A45F5/1516** (20250101); **A45F5/10** (20130101); **G06F1/1656** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of International Application No. PCT/KR2023/016409 designating the United States, filed on Oct. 20, 2023, in the Korean Intellectual Property Receiving Office and claiming priority to Korean Patent Application No. 10-2022-0162476, filed on Nov. 29, 2022, and Korean Patent Application No. 10-2022-0171924, filed on Dec. 9, 2022, the disclosures of which are all hereby incorporated by reference herein in their entireties.

TECHNICAL FIELD
[0002] Certain example embodiments may relate to an adapter accessory and/or an electronic device case including the same.
BACKGROUND ART
[0003] Recently, various types of electronic devices such as mobile phones, tablet personal computers (tablet PCs), MP3 players, portable multimedia players (PMPs), and e-books have been provided, allowing users to access various contents through these electronic devices. In addition to a wireless transmission and reception function, an electronic device is equipped with various functions such as a multimedia function such as photos, music, and video playback, and an entertainment function such as games. As mobile communication services have expanded into the multimedia service area, users have been able to use multimedia services as well as voice calls and short messages through electronic devices.
[0004] Electronic devices are made thin and light for portable purposes. Since such an electronic device is prone to damage due to external physical impact, it may be used with an external cover (or case) that covers the exterior of the electronic device. The external cover that covers the exterior of the electronic device may not only provide the function of protecting the electronic device, but also have the effect of evoking an aesthetic appeal to a user.
SUMMARY
[0005] According to an example embodiment, an adapter accessory may include an accessory body including a coupling portion configured to couple an external accessory thereto, a deformable portion extending from the accessory body and configured to be at least partially deformable by a force applied from an outside, a button portion connected, directly or indirectly, to the deformable portion and configured to press the deformable portion by the force applied from the outside or to be pressed by the deformable portion, and a fixing portion configured to be coupled, directly or indirectly, to an external case and formed to protrude from at least a portion of the button portion. A material of the deformable portion may be the same as a material of the button portion.
[0006] According to an example embodiment, an adapter accessory may include an accessory body including a coupling portion configured to couple an external accessory thereto, a deformable portion extending from the accessory body and configured to be at least partially deformable by a force applied from an outside, a button portion connected, directly or indirectly, to the deformable portion and configured to press the deformable portion by the force applied from the outside or to be pressed by the deformable portion, and a fixing portion configured to be coupled, directly or

indirectly, to an external case and formed to protrude from at least a portion of the button portion. The accessory body, the deformable portion, the button portion, and the fixing portion may be integrally formed.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] FIG. **1** is a perspective view illustrating an adapter accessory and a case which are separated from each other according to an example embodiment.

[0008] FIG. **2** is a perspective view illustrating an adapter accessory and a case which are coupled to each other according to an example embodiment.

[0009] FIG. **3** is a perspective view illustrating an adapter accessory according to an example embodiment.

[0010] FIG. **4** is a perspective view illustrating an adapter accessory according to an example embodiment.

[0011] FIG. **5**A is a cut-away perspective view illustrating an adapter accessory according to an example embodiment.

[0012] FIG. **5**B is a cross-sectional view illustrating an adapter accessory coupled to a case according to an example embodiment.

[0013] FIG. **5**C is a cross-sectional view illustrating an adapter accessory separated from a case according to an example embodiment.

[0014] FIG. **6**A is a cut-away perspective view illustrating an adapter accessory according to an example embodiment.

[0015] FIG. **6**B is a cross-sectional view illustrating an adapter accessory coupled to a case according to an example embodiment.

[0016] FIG. **6**C is a cross-sectional view illustrating an adapter accessory separated from a case according to an example embodiment.

[0017] FIG. **7**A is a cut-away perspective view illustrating an adapter accessory according to an example embodiment.

[0018] FIG. **7**B is a cross-sectional perspective view illustrating a portion of an adapter accessory according to an example embodiment.

[0019] FIG. **7**C is a cross-sectional view illustrating an adapter accessory coupled to a case according to an example embodiment.

[0020] FIG. **7**D is a cross-sectional view illustrating an adapter accessory separated from a case according to an example embodiment.

[0021] FIG. **8**A is a cut-away perspective view illustrating an adapter accessory according to an example embodiment.

[0022] FIG. **8**B is a cross-sectional perspective view illustrating a portion of an adapter accessory according to an example embodiment.

[0023] FIG. **8**C is a cross-sectional view illustrating an adapter accessory coupled to a case according to an example embodiment.

[0024] FIG. **8**D is a cross-sectional view illustrating an adapter accessory separated from a case according to an example embodiment.

[0025] FIG. **9**A is a cut-away perspective view illustrating an adapter accessory according to an example embodiment.

[0026] FIG. **9**B is a cross-sectional perspective view illustrating a portion of an adapter accessory according to an example embodiment.

[0027] FIG. **9**C is a cross-sectional view illustrating an adapter accessory coupled to a case according to an example embodiment.

[0028] FIG. **9**D is a cross-sectional view illustrating an adapter accessory separated from a case according to an example embodiment.
[0029] FIG. **10**A is a perspective view illustrating an adapter accessory according to an example embodiment.
[0030] FIG. **10**B is a cross-sectional view illustrating an adapter accessory according to an example embodiment.

DETAILED DESCRIPTION

[0031] It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as "A or B", "at least one of A and B", "at least one of A or B", "A, B, or C", "at least one of A, B, and C", and "at least one of A, B, or C", may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as "1.sup.st" and "2.sup.nd", or "first" and "second" may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term "operatively" or "communicatively", as "coupled with", "coupled to", "connected with", or "connected to" another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via at least a third element(s). Thus, for example, "connected" as used herein covers direct and indirect connections.
[0032] FIG. **1** is a perspective view illustrating an adapter accessory and a case which are separated from each other according to an example embodiment.
[0033] FIG. **2** is a perspective view illustrating an adapter accessory and a case which are coupled, directly or indirectly, to each other according to an example embodiment.
[0034] The embodiments of FIGS. **1** and **2** may be combined with the embodiments of FIGS. **3** to **10**B.
[0035] Referring to FIGS. **1** and **2**, an adapter accessory **100** may be detachably coupled, directly or indirectly, to a case **10** of an electronic device. In addition, an external accessory **20** may be detachably coupled to the adapter accessory **100**.
[0036] According to an embodiment, the case **10** of the electronic device may include a cover portion **11** surrounding at least a portion of the electronic device. The cover portion **11** may protect the electronic device from external impact (e.g., external impact such as chemical or physical impact) and provide various colors and textures to the appearance of the electronic device, thereby providing an aesthetic appeal to a user. The case **10** of the electronic device may also be referred to as an external case.
[0037] Various types of electronic devices may be mounted on the case **10** (or the cover portion **11**) of the electronic device. The electronic device may include at least one of, for example, a portable communication device (e.g., a smartphone), a tablet personal computer (PC), a mobile phone, a video phone, an e-book reader, a desktop PC, a laptop PC, a netbook computer, a workstation, a server, a personal digital assistant (PDA), a portable multimedia player (PMP), an MP3 player, a mobile medical device, a camera, or a wearable device. According to an embodiment, the electronic device mounted on the cover portion **11** may include any electronic device that is generally small enough to be carried by a user and allows a camera to be disposed on a front surface, a rear surface, or both surfaces thereof.
[0038] According to an embodiment, the cover portion **11** may include a flat surface that covers at least a portion of the electronic device (e.g., the rear surface of the electronic device) and a plurality

of side surfaces formed on edges of the flat surface. The plurality of side surfaces of the cover portion **11** may cover the side surfaces of the electronic device. The shape of the cover portion **11** illustrated in FIGS. **1** and **2** is exemplary, and may be subjected to various design modifications depending on the shape of the electronic device mounted on the case **10**.

[0039] According to an embodiment, the case **10** may further include at least one opening **13** formed on at least a portion of the flat surface of the cover portion **11**. The at least one opening **13** may accommodate a camera module (or camera deco member) of the electronic device, when the case **10** and the electronic device are coupled. The shape, position, or number of the at least one opening **13** in FIGS. **1** and **2** is exemplary, and may be provided in various shapes, positions, or numbers to accommodate the camera module or another exposed component of the electronic device.

[0040] According to an embodiment, the case **10** may further include a fastening portion **12** formed (or disposed) on the flat surface of the cover portion **11**. At least a portion of the adapter accessory **100** may be accommodated in the fastening portion **12**. Further, the adapter accessory **100** may be detachably fastened to the fastening portion **12**.

[0041] According to an embodiment, the fastening portion **12** may include a protrusion **12***a* formed to protrude from the plane of the cover portion **11**. The protrusion **12***a* may be formed in the form of a plurality of walls that are continuously connected to form a closed area. For example, the shape of the protrusion **12***a* may be an octagon as in the illustrated embodiment. However, the protrusion **12***a* is not limited thereto and may be formed in various polygonal shapes.

[0042] According to an embodiment, the fastening portion **12** may include at least one groove **12***b* formed to be recessed from at least a portion of the protrusion **12***a*. The at least one groove **12***b* may be formed on, directly or indirectly, each of a pair of inner walls of the protrusions **12***a* facing each other. A fixing portion (e.g., a fixing portion **140** of FIG. **4**) of the adapter accessory **100** may be inserted into the fastening portion **12**.

[0043] According to an embodiment, the adapter accessory **100** may be detachably coupled, directly or indirectly, to the fastening portion **12** of the case **10**. For example, the adapter accessory **100** may be fixed to the fastening portion **12** of the case **10** or detached from the fastening portion **12** of the case **10**. As described below, a user may easily couple the adapter accessory **100** to the case **10** or easily detach it from the case **10**. Accordingly, the adapter accessory **100** may be compatible with various cases including the fastening portion **12**. For example, as the single adapter accessory **100** is detachably coupled to each case including the fastening portion **12**, the user may use the single adapter accessory **100** for various cases.

[0044] According to an embodiment, the external accessory **20** may be detachably coupled to a coupling portion (e.g., a coupling portion **115** of FIG. **3**) of the adapter accessory **100**. For example, the external accessory **20** may be fixed to the coupling portion **115** of the adapter accessory **100** or detached from the coupling portion of the adapter accessory **100**. As described below, the user may easily couple the external accessory **20** to the adapter accessory **100** or easily detach it from the adapter accessory **100**. Accordingly, the user may couple various types of external accessories to the single adapter accessory **100**. For example, the user may couple various external accessories having a structure that may be coupled to the coupling portion of the adapter accessory **100** to the adapter accessory **100** and use them.

[0045] According to an exemplary embodiment, the illustrated external accessory **20** may be a scalable device (e.g., a pop-socket) that provides a grip function for the user to easily hold the electronic device or provides a stand function for the electronic device, to which the type of the external accessory **20** is not limited.

[0046] FIG. **3** is a perspective view illustrating an adapter accessory according to an example embodiment.

[0047] FIG. **4** is a perspective view illustrating the adapter accessory according to an example embodiment.

[0048] FIG. **4** is a perspective view illustrating the adapter accessory viewed from a different direction from in FIG. **3**.
[0049] The embodiments of FIGS. **3** and **4** may be combined with the embodiments of FIGS. **1** and **2** or the embodiments of FIGS. **5**A to **10**B.
[0050] Referring to FIGS. **3** and **4**, the adapter accessory **100** (e.g., the adapter accessory **100** of FIGS. **1** and **2**) may include an accessory body **110**, a deformable portion **120**, a button portion **130**, or the fixing portion **140**.
[0051] According to an embodiment, the accessory body **110** may be configured to cover at least a portion of a cover portion (e.g., the cover portion **11** of FIGS. **1** and **2**) of a case (e.g., the case **10** of FIGS. **1** and **2**) of an electronic device. The accessory body **110** may form, for example, a body of the adapter accessory **100**. The accessory body **110** may also be referred to as a housing, a body portion, or a cover portion.
[0052] According to an embodiment, the accessory body **110** may be formed in a circular shape as a whole. However, the accessory body **110** is not limited thereto, and may be formed in various shapes.
[0053] According to an embodiment, the accessory body **110** may include a first flat portion **111**, a first side portion **113**, the coupling portion **115**, a first support wall **117**, or a third support wall **118**.
[0054] According to an embodiment, when the adapter accessory **100** is coupled to the case, the first flat portion **111** may face the cover portion (e.g., the cover portion **11** of FIGS. **1** and **2**) of the case.
[0055] According to an embodiment, the first flat portion **111** may be formed in, but is not limited to, a circular shape. The first flat portion **111** may have a first surface (e.g., a surface facing a +Z direction in FIGS. **3** and **4**) facing a first direction (e.g., the +Z direction in FIGS. **3** and **4**) and a second surface (e.g., a surface facing a-Z direction in FIGS. **3** and **4**) facing in a direction (e.g., the −Z direction in FIGS. **3** and **4**) opposite to the first direction.
[0056] According to an embodiment, the first surface of the first flat portion **111** may be interpreted as a surface facing outward, when the adapter accessory **100** is coupled to the case. Further, the second surface of the first flat portion **111** may be interpreted as a surface facing the case, when the adapter accessory **100** is coupled to the case.
[0057] According to an embodiment, the first side portion **113** may be interpreted as a portion that is disposed along the edge of the first flat portion **111** and extends from the edge of the first flat portion **111** toward the case (e.g., the case **10** of FIGS. **1** and **2**). For example, the first side portion **113** may be disposed to surround a fastening portion (e.g., the fastening portion **12** of FIG. **1**) of the case, when the adapter accessory **100** is coupled to the case.
[0058] According to an embodiment, the first side portion **113** may extend from the edge of the first flat portion **111** in the second direction (e.g., the −Z direction of FIGS. **3** and **4**).
[0059] According to an embodiment, a portion where the first side portion **113** and the first flat portion **111** are connected may be formed to be curved, which should not be construed as limiting.
[0060] According to an embodiment, the coupling portion **115** may be formed on, directly or indirectly, the first surface of the first flat portion **111**. The coupling portion **115** may provide a space for coupling an external accessory (e.g., the external accessory **20** of FIGS. **1** and **2**) thereto. A detailed description of the coupling portion **115** will be described later.
[0061] According to an embodiment, the first support wall **117** and the third support wall **118** may be formed on, directly or indirectly, the second surface of the first flat portion **111**. The first support wall **117** may be configured to support at least a portion of the deformable portion **120**. The first support wall **117** may be formed to protrude from the second surface of the first flat portion **111** in the second direction (e.g., the −Z direction in FIGS. **3** and **4**). The first support wall **117** may be formed to extend in one direction.
[0062] According to an embodiment, the first support wall **117** may be disposed inside the first side portion **113**.

[0063] According to an embodiment, the third support wall **118** may extend from both ends of the first support wall **117**. For example, the third support wall **118** may extend from both ends of the first support wall **117** in an opposite direction to the deformable portion **120**. The third support wall **118** may support the first support wall **117**, so that the first support wall **117** is not bent when the deformable portion **120** is compressively deformed or restoratively deformed.
[0064] According to an embodiment, the first support wall **117** and the third support wall **118** may be disposed in an inner region formed by the first side portion **113**.
[0065] According to an embodiment, the deformable portion **120** may extend from the accessory body **110**. The deformable portion **120** may be configured to be at least partially deformable by a force applied from the outside.
[0066] According to an embodiment, the deformable portion **120** may extend from the first support wall **117** of the accessory body **110**. For example, the deformable portion **120** may extend from the first support wall **117** toward the outside of the accessory body **110** (e.g., an X-axis direction in FIG. **4**).
[0067] According to an embodiment, when the user presses the button portion **130**, the deformable portion **120** may be compressed by a force provided by the user, and when the user releases the pressure on the button portion **130**, the deformable portion **120** may be restored by an elastic restoring force.
[0068] According to an embodiment, the button portion **130** may be connected, directly or indirectly, to the deformable portion **120**. The button portion **130** may be configured to press the deformable portion **120** by an external force or to be pressed by an elastic restoring force of the deformable portion **120**.
[0069] According to an embodiment, when the user presses the button portion **130**, the button portion **130** may slide and compress the deformable portion **120**, and when the user releases the pressure on the button portion **130**, the button portion **130** may be pressed by the deformable portion **120** and slide. For example, when the user applies an external force to a pair of button portions **130**, deformable portions **120** may be compressed, and when the user releases the external force to the pair of button portions **130**, the deformable portions **120** may be restored.
[0070] According to an embodiment, the button portion **130** may include a second flat portion **131**, a second side portion **133**, a pattern portion **134**, or a second support wall **135**.
[0071] According to an embodiment, the button portion **130** may be disposed in a groove formed on at least one portion of a side surface of the accessory body **110**.
[0072] According to an embodiment, the second flat portion **131** of the button portion **130** may be disposed on substantially the same plane as the first flat portion **111** of the accessory body **110**.
[0073] According to an embodiment, the second side portion **133** may extend from an outer edge of the first flat portion **131** toward the cover portion of the case (e.g., the case **10** of FIGS. **1** and **2**). According to an embodiment, the second side portion **133** may be connected, directly or indirectly, to the edge of the first flat portion **131** so as to be angled.
[0074] According to an embodiment, the second side portion **133** may be formed with the pattern portion **134** including a plurality of patterns sunken from a surface of the second side portion **133**. The pattern portion **134** may be formed with a plurality of slit structures so that the second side portion **133** may have a rough surface. Accordingly, when the user presses the second side portion **133** of the button portion **130**, slipping of the user's hand may be prevented or reduced.
[0075] According to an embodiment, the second support wall **135** may extend from an inner edge of the second flat portion **131** toward the cover portion of the case. In an embodiment, the second support wall **135** may face the first support wall **117** with the deformable portion **120** interposed therebetween.
[0076] In an embodiment, when the user presses the button portion **130**, the second support wall **135** may press the deformable portion **120**, and when the user releases the pressure on the button portion **130**, the second support wall **135** may be pressed by the deformable portion **120**.

[0077] In an embodiment, at least a portion of the deformable portion **120** may be connected, directly or indirectly, to the second support wall **135**.
[0078] In an embodiment, the fixing portion **140** may extend from one end (e.g., an end facing the −Z direction in FIG. **4**) of the second support wall **135** toward the outside of the accessory body **110**. The fixing portion **140** may be interpreted as a portion of the button portion **130**.
[0079] According to an embodiment, the fixing portion **140** may be configured to be inserted into or removed from at least one groove (e.g., the at least one groove **12***b* of FIG. **1**) of the case (e.g., the case **10** of FIG. **1**).
[0080] According to an embodiment, the adapter accessory **100** may be formed of a non-metallic material (e.g., polycarbonate). Further, the deformable portion **120**, the button portion **130**, or the fixing portion **140** of the adapter accessory **100** may be formed integrally.
[0081] According to an embodiment, the accessory body **110**, the deformable portion **120**, the button portion **130**, or the fixing portion **140** of the adapter accessory **100** may be formed of substantially the same material. For example, the accessory body **110**, the deformable portion **120**, the button portion **130**, or the fixing portion **140** may be formed of a non-metallic material (e.g., polycarbonate).
[0082] According to an embodiment, the accessory body **110**, the deformable portion **120**, the button portion **130**, or the fixing portion **140** of the adapter accessory **100** may include a different material (e.g., a metal material).
[0083] According to an embodiment, a pair of button portions **130** may be provided. For example, the pair of button portions **130** may be disposed to face each other. According to an embodiment, a pair of deformable portions **120**, a pair of first support walls **117**, and a pair of third support walls **118** may be provided to correspond to the pair of button portions **130** in the adapter accessory **100**.
[0084] FIG. **5**A is a cut-away perspective view illustrating an adapter accessory according to an example embodiment.
[0085] FIG. **5**B is a cross-sectional view illustrating an adapter accessory coupled to a case according to an example embodiment.
[0086] FIG. **5**C is a cross-sectional view illustrating an adapter accessory separated from a case according to an example embodiment.
[0087] The embodiments of FIGS. **5**A to **5**C may be combined with the embodiments of FIGS. **1** to **4** or the embodiments of FIGS. **6**A to **10**B.
[0088] According to an embodiment, the coupling portion **115** may include a bottom portion **115***a*, a protruding portion **115***b*, and a loop portion **115***c*.
[0089] According to an embodiment, the bottom portion **115***a* may form substantially the same plane as the first flat portion **111**. For example, the bottom portion **115***a* may be interpreted as a portion of the first flat portion **111**.
[0090] According to an embodiment, the protruding portion **115***b* may be formed in a substantially circular shape, and may be a portion protruding in the first direction (e.g., the +Z direction in FIG. **5**A) from the first flat portion **111** or the bottom portion **115***a*.
[0091] According to an embodiment, the loop portion **115***c* may be a portion that extends inward from a top end (e.g., an end facing the +Z direction in FIG. **5**A) of the protruding portion **115***b*. The loop portion **115***c* may be formed only on a portion of the protruding portion **115***b*, not on the remaining portion of the protruding portion **115***b*.
[0092] According to an embodiment, a fixing portion of the external accessory (e.g., the external accessory **20** of FIGS. **1** and **2**) may be inserted into the inside of the protruding portion **115***b* through the portion where the loop portion **115***c* is not formed. Then, when the external accessory is rotated, the fixing portion of the external accessory is disposed between the loop portion **115***c* and the bottom portion **115***a*, thereby preventing or reducing chances of the external accessory from being separated from the coupling portion **115**.
[0093] According to an embodiment, the deformable portion **120** of the adapter accessory **100** may

include a first deformable part **121** or a second deformable part **123**.

[0094] According to an embodiment, the first deformable part **121** may connect a top end (e.g., an end facing the +Z direction in FIG. **5**A) of the first support wall **117** and a top end (e.g., an end facing the +Z direction in FIG. **5**A) of the second support wall **135**.

[0095] According to an embodiment, the second deformable part **123** may connect a bottom end (e.g., an end facing the −Z direction in FIG. **5**A) of the first support wall **117** and a bottom end (e.g., an end facing the −Z direction in FIG. **5**A) of the second support wall **135**.

[0096] According to an embodiment, each of the first deformable part **121** and the second deformable part **123** may be formed to be at least partially bendable.

[0097] According to an embodiment, the first deformable part **121** may include a first inclined part **121***a* and a second inclined part **121***b*. The first inclined part **121***a* may extend from the top end of the first support wall **117**. Further, the first inclined part **121***a* may be inclined from the first support wall **117** to form an acute angle with respect to the first support wall **117**. For example, the first inclined part **121***a* may be interpreted as extending from the top end of the first support wall **117** toward the bottom end of the second support wall **135**.

[0098] According to an embodiment, the first inclined part **121***a* may be connected, directly or indirectly, to the second inclined part **121***b*.

[0099] The second inclined part **121***b* may extend from the top end of the second support wall **135**. Further, the second inclined part **121***b* may be inclined from the second support wall **135** to form an acute angle with respect to the second support wall **135**. For example, the second inclined part **121***b* may be interpreted as extending from the top end of the second support wall **135** toward the bottom end of the first support wall **117**.

[0100] According to an embodiment, the second inclined part **121***b* may be connected, directly or indirectly, to the first inclined part **121***a*.

[0101] According to an embodiment, the first inclined part **121***a* and the second inclined part **121***b* may be connected in a ‘V’ shape, and the vertex of the ‘V’ shape may be disposed to face the second deformable part **123**.

[0102] According to an embodiment, the second deformable part **123** may include a third inclined part **123***a* and a fourth inclined part **123***b*. The third inclined part **123***a* may extend from the bottom end of the first support wall **117**. Further, the third inclined part **123***a* may be inclined from the first support wall **117** to form an acute angle with respect to the first support wall **117**. For example, the third inclined part **123***a* may be interpreted as extending from the bottom end of the first support wall **117** toward the top end of the second support wall **135**.

[0103] According to an embodiment, the third inclined part **123***a* may be connected, directly or indirectly, to the fourth inclined part **123***b*.

[0104] The fourth inclined part **123***b* may extend from the bottom end of the second support wall **135**. Further, the fourth inclined part **123***b* may be inclined from the second support wall **135** to form an acute angle with respect to the second support wall **135**. For example, the fourth inclined part **123***b* may be interpreted as extending from the bottom end of the second support wall **135** toward the top end of the first support wall **117**.

[0105] According to an embodiment, the fourth inclined part **123***b* may be connected, directly or indirectly, to the third inclined part **123***a*.

[0106] According to an embodiment, the third inclined part **123***a* and the fourth inclined part **123***b* may be connected in a ‘V’ shape, and the vertex of the ‘V’ shape may be disposed to face the first deformable part **121**.

[0107] Referring to FIG. **5**B, a pair of fixing portions **140** of the adapter accessory **100** may be inserted into the at least one groove **12***b* of the case **10** of the electronic device.

[0108] Accordingly, since the pair of fixed portions **140** are fixedly caught in at least a portion of the protrusion **12***a* of the case **10**, the adapter accessory **100** may be prevented, or chances reduced, from being separated from the case **10**.

[0109] Referring to FIG. **5**C, the deformable portion **120** is shown as compressed. For example, when the user presses the button portion **130** (e.g., the second side portion **135**) (e.g., applies a force in a direction P), the deformable portion **120** may be compressed by the button portion **130**, while at least a portion of the deformable portion **120** is bent. When the deformable portion **120** is compressed, the first support wall **117** and the second support wall **135** may come closer.
[0110] According to an embodiment, when the deformable portion **120** is compressed, the fixing portion **140** connected to the button portion **130** may be separated from the at least one groove **12***b*. As illustrated in FIG. **5**C, with the fixing portion **140** detached from the at least one groove **12***b*, the adapter accessory **100** may be separated from the case **10**.
[0111] In order to couple the adapter accessory **100** to the case **10**, the deformable portion **120** may be compressed by pressing the button portion **130**, and then the adapter accessory **100** may be seated on the fastening portion **12** of the case **10**. Further, with the adapter accessory **100** seated on the fastening portion **12**, when the pressure on the button portion **130** is released, the deformable portion **120** may be elastically restored, so that the fixing portion **140** may be inserted into the at least one groove **12***b*, as illustrated in FIG. **5**B.
[0112] According to an embodiment, the first inclined part **121***a*, the second inclined part **121***b*, the third inclined part **123***a*, and the fourth inclined part **123***b* may have substantially the same thickness d**3**.
[0113] According to an embodiment, the thickness d**3** of the first inclined part **121***a*, the second inclined part **121***b*, the third inclined part **123***a*, and the fourth inclined part **123***b* may be smaller than the thickness d**1** of the first support wall **117**. In addition, the thickness d**3** of the first inclined part **121***a*, the second inclined part **121***b*, the third inclined part **123***a*, and the fourth inclined part **123***b* may be smaller than the thickness d**2** of the second support wall **135**. Accordingly, when the deformable portion **120** is compressed or elastically restored, the bending of the first support wall **117** or the second support wall **135** may be prevented or reduced.
[0114] According to an embodiment, the deformable portion **120** may include at least one recess **125**. When each of the components of the deformable portion **120** is folded or unfolded relative to another component, the at least one recess **125** may prevent or reduce a chance of a portion connected to the other component from being torn. For example, the at least one recess **125** may be defined as a circular recess structure or a circular pivot structure.
[0115] According to an embodiment, the at least one recess **125** may be formed to be sunken further in at least a portion of the first deformable part **121** or at least a portion of the second deformable part **123** than in the other portion thereof.
[0116] According to an embodiment, the fixing portion **140** may include an inclined surface **140***a* to form a first angle g with respect to the second support wall **135**. The first angle g may be an obtuse angle. For example, the first angle g may be an angle greater than about 90 degrees and less than about 135 degrees. According to an embodiment, since the inclined surface **140***a* of the fixing portion **140** is formed at the obtuse first angle g, even if the button portion **130** is twisted within a predetermined range when the user presses the button portion **130**, the fixing portion **140** may be easily inserted into the at least one groove **12***b*.
[0117] FIG. **6**A is a cut-away perspective view illustrating an adapter accessory according to an example embodiment.
[0118] FIG. **6**B is a cross-sectional view illustrating an adapter accessory coupled to a case according to an example embodiment.
[0119] FIG. **6**C is a cross-sectional view illustrating an adapter accessory separated from a case according to an example embodiment.
[0120] The embodiments of FIGS. **6**A to **6**C may be combined with the embodiments of FIGS. **1** to **5**C or the embodiments of FIGS. **7**A to **10**B.
[0121] According to an embodiment, the deformable portion **120** of the adapter accessory **100** may include a first deformable part **221** or a second deformable part **223**.

[0122] According to an embodiment, the first deformable part **221** may connect the top end (e.g., an end facing the +Z direction in FIG. **6**A) of the first support wall **117** and the top end (e.g., an end facing the +Z direction in FIG. **6**A) of the second support wall **135**.
[0123] According to an embodiment, the second deformable part **223** may connect the bottom end (e.g., an end facing the −Z direction in FIG. **6**A) of the first support wall **117** and the bottom end (e.g., an end facing the −Z direction in FIG. **6**A) of the second support wall **135**.
[0124] According to an embodiment, each of the first deformable part **221** and the second deformable part **223** may be formed to be at least partially bendable.
[0125] According to an embodiment, the first deformable part **221** may include a first inclined part **221***a* and a second inclined part **221***b*. The first inclined part **221***a* may extend from a portion of the first support wall **117**. For example, the first inclined part **221***a* may be connected to a point between the top end (e.g., the end facing the +Z direction in FIG. **6**A) and the bottom end (e.g., the end facing the −Z direction in FIG. **6**A) of the first support wall **117**. Further, the first inclined part **221***a* may be inclined from the first support wall **117** to form an acute angle with respect to the first support wall **117**. For example, the first inclined part **221***a* may be interpreted as extending from a point of the first support wall **117** toward the top end of the second support wall **135**.
[0126] According to an embodiment, the first inclined part **221***a* may be connected to the second inclined part **221***b*.
[0127] The second inclined part **221***b* may extend from a portion of the second support wall **135**. For example, the second inclined part **221***b* may be connected to a point between the top end (e.g., an end facing the +Z direction in FIG. **6**A) and the bottom end (e.g., an end facing the −Z direction in FIG. **6**A) of the second support wall **135**. Further, the second inclined part **221***b* may be inclined from the second support wall **135** to form an acute angle with respect to the second support wall **135**. For example, the second inclined part **221***b* may be interpreted as extending from a point of the second support wall **135** toward the top end of the first support wall **117**.
[0128] According to an embodiment, the second inclined part **221***b* may be connected to the first inclined part **221***a*.
[0129] According to an embodiment, the first inclined part **221***a* and the second inclined part **221***b* may be connected in a ‘V’ shape, and the vertex of the ‘V’ shape of the first inclined part **221***a* and the second inclined part **221***b* may face in an opposite direction to the vertex of a ‘V’ shape of a third inclined part **223***a* and a fourth inclined part **223***b*.
[0130] According to an embodiment, the second deformable part **223** may include the third inclined part **223***a* and the fourth inclined part **223***b*. The third inclined part **223***a* may extend from a portion of the first support wall **117**. For example, the third inclined part **223***a* may be connected to a point between the top end (e.g., the end facing the +Z direction in FIG. **6**A) and the bottom end (e.g., the end facing the −Z direction in FIG. **6**A) of the first support wall **117**. Further, the third inclined part **223***a* may be inclined from the first support wall **117** to form an acute angle with respect to the first support wall **117**. For example, the third inclined part **223***a* may be interpreted as extending from a point of the first support wall **117** toward the bottom end of the second support wall **135**.
[0131] According to an embodiment, the third inclined part **223***a* may be connected to the fourth inclined part **223***b*.
[0132] The fourth inclined part **223***b* may extend from a portion of the second support wall **135**. For example, the fourth inclined part **223***b* may be connected to a point between the top end (e.g., the end facing the +Z direction in FIG. **6**A) and the bottom end (e.g., the end facing the −Z direction in FIG. **6**A) of the second support wall **135**. Further, the fourth inclined part **223***b* may be inclined from the second support wall **135** to form an acute angle with respect to the second support wall **117**. For example, the fourth inclined part **223***b* may be interpreted as extending from a point of the second support wall **135** toward the bottom end of the first support wall **117**.
[0133] According to an embodiment, the fourth inclined part **223***b* may be connected to the third inclined part **223***a*.

[0134] According to an embodiment, the third inclined part **223***a* and the fourth inclined part **223***b* may be connected in the 'V' shape, and the vertex of the 'V' shape of the third inclined part **223***a* and the fourth inclined part **223***b* may face in an opposite direction to the vertex of the 'V' shape of the first inclined part **221***a* and the second inclined part **221***b*.

[0135] Referring to FIG. **6**B, a pair of fixing portions **140** of the adapter accessory **100** may be inserted into the at least one groove **12***b* of the case **10** of the electronic device. Accordingly, as the pair of fixing portions **140** are fixedly caught in at least a portion of the protrusion **12***a* of the case **10**, the adapter accessory **100** may be prevented or reduced from being separated from the case **10**.

[0136] Referring to FIG. **6**C, the deformable portion **120** is shown as compressed. For example, when the user presses the button portion **130** (e.g., the second side portion **135**) (e.g., applies a force in the direction P), the deformable portion **120** may be compressed by the button portion **130**, while at least a portion of the deformable portion **120** is bent. When the deformable portion **120** is compressed, the first support wall **117** and the second support wall **135** may come closer.

[0137] According to an embodiment, when the deformable portion **120** is compressed, the fixing portion **140** connected to the button portion **130** may be separated from the at least one groove **12***b*. As illustrated in FIG. **6**C, with the fixing portion **140** detached from the at least one groove **12***b*, the adapter accessory **100** may be separated from the case **10**.

[0138] In order to couple the adapter accessory **100** to the case **10**, the deformable portion **120** may be compressed by pressing the button portion **130**, and then the adapter accessory **100** may be seated on the fastening portion **12** of the case **10**. Further, when the button portion **130** is released with the adapter accessory **100** seated on the fastening portion **12**, the deformable portion **120** may be elastically restored, and thus the fixing portion **140** may be inserted into the at least one groove **12***b*, as illustrated in FIG. **6**B.

[0139] According to an embodiment, the first inclined part **221***a*, the second inclined part **221***b*, the third inclined part **223***a*, and the fourth inclined part **223***b* may have substantially the same thickness d**3**.

[0140] According to an embodiment, the thickness d**3** of the first inclined part **221***a*, the second inclined part **221***b*, the third inclined part **223***a*, and the fourth inclined part **223***b* may be smaller than the thickness d**1** of the first support wall **117**. Further, the thickness d**3** of the first inclined part **221***a*, the second inclined part **221***b*, the third inclined part **223***a*, and the fourth inclined part **223***b* may be smaller than the thickness d**2** of the second support wall **135**. Accordingly, when the deformable portion **120** is compressed or elastically restored, the bending of the first support wall **117** or the second support wall **135** may be prevented or reduced.

[0141] According to an embodiment, the deformable portion **120** may include at least one recess **225**. When each of the components of the deformable portion **120** is folded or unfolded relative to another component, the at least one recess **225** may prevent or reduce a chance of a portion connected to the other component from being torn. For example, the at least one recess **225** may be defined as a circular recess structure or a circular pivot structure.

[0142] According to an embodiment, the at least one recess **225** may be formed to be sunken further in at least a portion of the first deformable part **221** or at least a portion of the second deformable part **223** than the other portion thereof.

[0143] FIG. **7**A is a cut-away perspective view illustrating an adapter accessory according to an example embodiment.

[0144] FIG. **7**B is a cross-sectional perspective view illustrating a portion of an adapter accessory according to an example embodiment.

[0145] FIG. **7**C is a cross-sectional view illustrating an adapter accessory coupled to a case according to an example embodiment.

[0146] FIG. **7**D is a cross-sectional view illustrating an adapter accessory separated from a case according to an example embodiment.

[0147] The embodiments of FIGS. **7**A to **7**D may be combined with the embodiments of FIGS. **1** to

6C or the embodiments of FIGS. **8**A to **10**B.

[0148] According to an embodiment, the deformable portion **120** of the adapter accessory **100** may include a first deformable part **321**.

[0149] According to an embodiment, the first deformable part **321** may connect the top end (e.g., an end facing the +Z direction in FIG. **7**A) of the first support wall **117** and the bottom end (e.g., an end facing the −Z direction in FIG. **7**A) of the second support wall **135**.

[0150] According to an embodiment, the first deformable part **321** may be formed to be at least partially bendable. For example, with respect to a portion of the first deformable part **321** connected to the first support wall **117**, the first deformable part **321** may be folded or unfolded relative to the first support wall **117**. Further, with respect to a portion of the first deformable part **321** connected to the second support wall **135**, the first deformable part **321** may be folded or unfolded relative to the second support wall **135**.

[0151] According to an embodiment, the first deformable part **321** may include at least one first opening **322** formed to penetrate at least one portion thereof. The first opening **322** may be interpreted as a portion formed by cutting through at least one portion of the first deformable part **321**.

[0152] According to an embodiment, since the first opening **322** is formed in the first deformable part **321**, when the first deformable part **321** is compressively deformed or elastically restored, the first deformable part **321** may be more easily elastically deformed.

[0153] While the deformable portion **120** is not compressed (e.g., FIG. **7**C), the first support wall **117** and the second support wall **135** may be spaced apart from each other by a first distance L**1**. While the deformable portion **120** is compressed (e.g., FIG. **7**D), the first support wall **117** and the second support wall **135** may be spaced apart from each other by a second distance L**2** smaller than the first distance L**1**. For example, the distance L**2** between the first support wall **117** and the second support wall **135** in the compressed state (e.g., FIG. **7**D) of the deformable portion **120** may be smaller than the distance L**1** between the first support wall **117** and the second support wall **135** in the elastically restored state (e.g., FIG. **7**C).

[0154] Referring to FIG. **7**C, a pair of fixing portions **140** of the adapter accessory **100** may be inserted into the at least one groove **12**B of the case **10** of the electronic device. Accordingly, as the pair of fixing portions **140** are fixedly caught in at least a portion of the protrusion **12***a* of the case **10**, the adapter accessory **100** may be prevented from being separated from the case **10**.

[0155] Referring to FIG. **7**D, the deformable portion **120** is shown as compressed. For example, when the user presses the button portion **130** (e.g., the second side portion **135**) (e.g., applies a force in the direction P), the deformable portion **120** may be compressed. When the deformable portion **120** is compressed, the first support wall **117** and the second support wall **135** may come closer at the second distance L**2**.

[0156] According to an embodiment, while the deformable portion **120** is compressed, the fixing portion **140** connected to the button portion **130** may be separated from the at least one groove **12***b*. As illustrated in FIG. **7**D, with the fixing portion **140** detached from the at least one groove **12***b*, the adapter accessory **100** may be separated from the case **10**.

[0157] In order to couple the adapter accessory **100** to the case **10**, the deformable portion **120** may be compressed by pressing the button portion **130**, and then the adapter accessory **100** may be seated on the fastening portion **12** of the case **10**. Further, when the pressure on the button portion **130** is released with the adapter accessory **100** seated on the fastening portion **12**, the deformable portion **120** may be elastically restored, and thus the fixing portion **140** may be inserted into the at least one groove **12***b*, as illustrated in FIG. **7**C.

[0158] According to an embodiment, the deformable portion **120** may include at least one recess **325**. The at least one recess **325** may prevent or reduce a chance of a portion connected to the first support wall **117** and/or the second support wall **135** from being torn, when the first deformable part **321** of the deformable portion **120** is folded or unfolded relative to the first support wall **117** or the second support wall **135**. For example, the at least one recess **325** may be defined as a circular recess structure or a circular pivot structure.

[0159] According to an embodiment, the at least one recess **325** may be formed to be sunken further in at least one portion of the first deformable part **321** than in the other portion thereof.

[0160] FIG. **8**A is a cut-away perspective view illustrating an adapter accessory according to an example embodiment.

[0161] FIG. **8**B is a cross-sectional perspective view illustrating a portion of an adapter accessory according to an example embodiment.

[0162] FIG. **8**C is a cross-sectional view illustrating an adapter accessory coupled to a case according to an example embodiment.

[0163] FIG. **8**D is a cross-sectional view illustrating an adapter accessory separated from a case according to an example embodiment.

[0164] The embodiments of FIGS. **8**A to **8**D may be combined with the embodiments of FIGS. **1** to **7**D or the embodiments of FIGS. **9**A to **10**B.

[0165] According to an embodiment, the deformable portion **120** of the adapter accessory **100** may include a first deformable part **421** or a second deformable part **423**.

[0166] According to an embodiment, the first deformable part **421** may connect the bottom end (e.g., an end facing the −Z direction in FIG. **8**A) of the second support wall **135** and at least a portion of the first flat portion **111**.

[0167] According to an embodiment, the first deformable part **421** may be formed to be at least partially bendable.

[0168] According to an embodiment, the first deformable part **421** may include at least one first opening **422** formed to penetrate at least one portion thereof. The first opening **422** may be interpreted as a portion formed by cutting through at least one portion of the first deformable part **421**.

[0169] According to an embodiment, when the first opening **422** is formed in the first deformable part **421**, the first deformable part **421** may be more easily elastically deformed, when the first deformable part **421** is compressively deformed or elastically restored.

[0170] According to an embodiment, at least a portion of the second deformable part **423** may be accommodated in the first opening **422**.

[0171] According to an embodiment, the second deformable part **423** may include a first inclined part **423***a* and a second inclined part **423***b*. The first inclined part **423***a* may extend from a portion of the first flat portion **111**. Further, the first inclined part **423***a* may extend to be inclined with respect to the first flat portion **111**. For example, the first inclined part **423***a* may be interpreted as extending from a point (or a portion) of the first flat portion **111** toward the bottom end (e.g., an end facing the −Z direction in FIG. **8**A) of the second support wall **135**.

[0172] According to an embodiment, the first inclined part **423***a* may be connected to the second inclined part **423***b*.

[0173] According to an embodiment, the second inclined part **423***b* may extend from the top end (e.g., an end facing the +Z direction in FIG. **6**A) of the second support wall **135**. Further, the second inclined part **423***b* may be inclined from the second support wall **135** to form an acute angle with respect to the second support wall **135**.

[0174] According to an embodiment, the second support wall **135** may be formed to have a second opening **136** penetrating at least a portion of the second support wall **135**. At least a portion of the second deformable part **423** may be accommodated in the second opening **136**.

[0175] Referring to FIG. **8**C, a pair of fixing portions **140** of the adapter accessory **100** may be inserted into the at least one groove **12***b* of the case **10** of the electronic device. Accordingly, since the pair of fixed portions **140** are fixedly caught in at least a portion of the protrusion **12***a* of the case **10**, the adapter accessory **100** may be prevented or reduced from being separated from the case **10**.

[0176] Referring to FIG. **8**D, the deformable portion **120** is shown as compressed. For example, when the user presses the button portion **130** (e.g., the second side portion **135**) (e.g., applies a force in the direction P), the deformable portion **120** may be compressed. When the deformable portion **120** is compressed, an acute angle formed by the first inclined part **423***a* and the second inclined part **423***b* may be reduced. For example, the acute angle formed by the first inclined part **423***a* and the second inclined part **423***b* in the compressed state (e.g., FIG. **8**D) of the deformable portion **120** may be smaller than the acute angle formed by the first inclined part **423***a* and the second inclined part **423***b* in the non-compressed state (e.g., FIG. **8**C) of the deformable portion **120**.
[0177] According to an embodiment, while the deformable portion **120** is compressed, the fixing portion **140** connected to the button portion **130** may be detached from the at least one groove **12***b*. As illustrated in FIG. **8**D, with the fixing portion **140** detached from at least one groove **12***b*, the adapter accessory **100** may be separated from the case **10**.
[0178] In order to couple the adapter accessory **100** to the case **10**, the deformable portion **120** may be compressed by pressing the button portion **130**, and then the adapter accessory **100** may be seated on the fastening portion **12** of the case **10**. Further, when the pressure on the button portion **130** is released with the adapter accessory **100** seated on the fastening portion **12**, the deformable portion **120** may be elastically restored, and thus the fixing portion **140** may be inserted into the at least one groove **12***b*, as illustrated in FIG. **8**C.
[0179] According to an embodiment, the deformable portion **120** may include at least one recess **425**. When the first inclined part **423***a* and the second inclined part **423***b* are folded or unfolded relative to each other, the at least one recess **425** may prevent or reduce a chance of a portion where the first inclined part **423***a* and the second inclined part **423***b* are connected from being torn. For example, the at least one recess **425** may be defined as a circular recess structure or a circular pivot structure.
[0180] According to an embodiment, the at least one recess **425** may be formed to be sunken further in at least one portion of the second deformable part **423** than in the other portion thereof.
[0181] FIG. **9**A is a cut-away perspective view illustrating an adapter accessory according to an example embodiment.
[0182] FIG. **9**B is a cross-sectional perspective view illustrating a portion of an adapter accessory according to an example embodiment.
[0183] FIG. **9**C is a cross-sectional view illustrating an adapter accessory coupled to a case according to an example embodiment.
[0184] FIG. **9**D is a cross-sectional view illustrating an adapter accessory separated from a case according to an example embodiment.
[0185] The embodiments of FIGS. **9**A to **9**D may be combined with the embodiments of FIGS. **1** to **8**D or the embodiments of FIGS. **10**A and **10**B.
[0186] According to an embodiment, the deformable portion **120** of the adapter accessory **100** may include a first deformable part **521** or a second deformable part **523**.
[0187] According to an embodiment, the first deformable part **521** may connect at least a portion of the first flat portion **111** and at least a portion of the second flat portion **131**.
[0188] According to an embodiment, the first deformable part **521** may be formed to be at least partially bendable.
[0189] According to an embodiment, the first deformable part **521** may include a first inclined part **521***a* and a second inclined part **521***b*.
[0190] The first inclined part **521***a* may extend from a portion of the first flat portion **111**. Further, the first inclined part **521***a* may extend to be inclined with respect to the first flat portion **111**.
[0191] According to an embodiment, the first inclined part **521***a* may be connected to the second inclined part **521***b*.
[0192] The second inclined part **521***b* may extend from a portion of the second flat portion **131**.

Further, the second inclined part **521**_b_ may extend to be inclined with respect to the second flat portion **131**.

[0193] According to an embodiment, the second inclined part **521**_b_ may be connected to the first inclined part **521**_b_.

[0194] According to an embodiment, the first inclined part **521**_a_ may include a first opening **521**_a_-**1** formed to penetrate at least a portion thereof. The first opening **521**_a_-**1** may be interpreted as a portion formed by cutting through at least a portion of the first inclined part **521**_a_.

[0195] According to an embodiment, at least a portion of the second deformable part **523** (e.g., a third inclined part **523**_a_ or a fourth inclined part **523**_b_) may be accommodated in the first opening **521**_a_-**1**.

[0196] According to an embodiment, the second inclined part **521**_b_ may include a second opening **521**_b_-**1** formed to penetrate at least a portion thereof. The second opening **521**_b_-**1** may be interpreted as a portion formed by cutting through at least a portion of the second inclined part **521**_b_.

[0197] According to an embodiment, at least a portion (e.g., a fifth inclined part **523**_c_ or a sixth inclined part **523**_d_) of the second deformable part **523** may be accommodated in the second opening **521**_b_-**1**.

[0198] According to an embodiment, since the first opening **521**_a_-**1** and the second opening **521**_b_-**1** are formed in the first deformable part **521**, when the first deformable part **521** is compressively deformed or elastically restored, the first deformable part **521** may be elastically deformed more easily.

[0199] According to an embodiment, the second deformable part **523** may include the third inclined part **523**_a_, the fourth inclined part **523**_b_, the fifth inclined part **523**_c_, or the sixth inclined part **523**_d_.

[0200] The third inclined part **523**_a_ may extend to be inclined from at least a portion of the first flat portion **111**. The fourth inclined part **523**_b_ may be connected to the third inclined part **523**_a_ so as to form an acute angle with respect to the third inclined part **523**_a_.

[0201] The fifth inclined part **523**_c_ may be connected to the fourth inclined part **523**_b_ so as to form an acute angle with respect to the fourth inclined part **523**_b_. The sixth inclined part **523**_d_ may be connected to the fifth inclined part **523**_c_ so as to form an acute angle with respect to the fifth inclined part **523**_c_. The sixth inclined part **523**_d_ may extend to be inclined from at least a portion of the second flat portion **131**.

[0202] According to an embodiment, the third inclined part **523**_a_, the fourth inclined part **523**_b_, the fifth inclined part **523**_c_, and the sixth inclined part **523**_d_ may be connected in, but are not limited to, a ‘W’ shape.

[0203] At least a portion of the third inclined part **523**_a_ and the fourth inclined part **523**_b_ may be accommodated in the first opening **521**_a_-**1**. At least a portion of the fifth inclined part **523**_c_ and the sixth inclined part **523**_d_ may be accommodated in the second opening **521**_b_-**1**.

[0204] Referring to FIG. **9**C, a pair of fixing portions **140** of the adapter accessory **100** may be inserted into the at least one groove **12**_b_ of the case **10** of the electronic device. Accordingly, since the pair of fixed portions **140** are fixedly caught in at least a portion of the protrusion **12**_a_ of the case **10**, the adapter accessory **100** may be prevented or reduced from being separated from the case **10**.

[0205] Referring to FIG. **9**D, the deformable portion **120** is shown as compressed. For example, when the user presses the button portion **130** (e.g., the second side portion **135**) (e.g., applies a force in the direction P), the deformable portion **120** may be compressed. When the deformable portion **120** is compressed, an acute angle B**2** formed by the fourth inclined part **523**_b_ and the fifth inclined part **523**_c_ may be reduced. For example, the acute angle B**2** formed by the fourth inclined part **523**_b_ and the fifth inclined part **523**_c_ may be smaller in the compressed state (e.g., FIG. **9**D) of the deformable portion **120** than an acute angle B**1** formed by the fourth inclined part **523**_b_ and the fifth inclined part **523**_c_ in the non-compressed state (e.g., FIG. **9**C) of the deformable portion **120**.

[0206] According to an embodiment, while the deformable portion **120** is compressed, the fixing portion **140** connected to the button portion **130** may be detached from the at least one groove **12***b*. As illustrated in FIG. **9**D, with the fixing portion **140** detached from at least one groove **12***b*, the adapter accessory **100** may be separated from the case **10**.

[0207] In order to couple the adapter accessory **100** to the case **10**, the deformable portion **120** may be compressed by pressing the button portion **130**, and then the adapter accessory **100** may be seated on the fastening portion **12** of the case **10**. Further, with the adapter accessory **100** seated on the fastening portion **12**, when the pressure on the button portion **130** is released, the deformable portion **120** may be elastically restored, and thus the fixing portion **140** may be inserted into the at least one groove **12***b*, as illustrated in FIG. **9**C.

[0208] According to an embodiment, the deformable portion **120** may include at least one recess **525**. The at least one recess **525** may include a first recess **525***a* that prevents or reduces chances of a portion where the first inclined part **523***a* and the second inclined part **523***b* are connected from being torn, when the first inclined part **521***a* and the second inclined part **521***b* are folded or unfolded relative to each other. For example, the at least one first recess **525***a* may be defined as a circular recess structure or a circular pivot structure. According to an embodiment, at least one first recess **525***a* may be formed to be sunken further in at least one portion of the first deformable part **521** than in the other portion thereof.

[0209] The at least one recess **525** may include a second recess **525***b* that prevents a portion connected to another component from being torn, when a component of the second deformable part **523** is folded or unfolded relative to the other component. For example, at least one second recess **525***b* may be defined as a circular recess structure or a circular pivot structure. In an embodiment, the at least one second recess **525***b* may be formed to be sunken further in at least one portion of the second deformable part **523** than in the other portion thereof.

[0210] FIG. **10**A is a perspective view illustrating an adapter accessory according to an example embodiment.

[0211] FIG. **10**B is a cross-sectional view illustrating an adapter accessory according to an example embodiment.

[0212] The embodiments of FIGS. **10**A and **10**B may be combined with the embodiments of FIGS. **1** to **9**D.

[0213] Referring to FIGS. **10**A and **10**B, the adapter accessory **100** (e.g., the adapter accessory **100** of FIGS. **1** and **2**) may include the accessory body **110**, the deformable portion **120**, the button portion **130**, or the fixing portion **140**.

[0214] According to an embodiment, the accessory body **110** may include the first flat portion **111**, the first side portion **113**, or the coupling portion **115**.

[0215] According to an embodiment, the button portion **130** may be connected to the deformable portion **120**. The button portion **130** may be configured to press the deformable portion **120** by a force applied from the outside or to be pressed by the deformable portion **120**.

[0216] According to an embodiment, when the user presses the button portion **130**, the button portion **130** may slide and compress the deformable portion **120**, and when the user releases the pressure on the button portion **130**, the button portion **130** may be compressed by the deformable portion **120** and slide.

[0217] According to an embodiment, the button portion **130** may include the second flat portion **131**, a second side portion **233**, a curved portion **232**, or the second support wall **135**.

[0218] According to an embodiment, the curved portion **232** may be a portion connecting an edge of the second flat portion **131** and the second side portion **233**. For example, the curved portion **232** may include a curved surface so that the portion where the second flat portion **131** and the second side portion **233** are connected is in a rounded shape.

[0219] According to an embodiment, the deformable portion **120** may include a first deformable part **621**. The first deformable part **621** may extend from at least a portion of the first flat portion

111. The first deformable part **621** may extend to be inclined with respect to the first flat portion **111**. The first deformable part **621** may extend toward the bottom end (e.g., an end facing the −Z direction in FIG. **10**A) of the second support wall **135**.

[0220] An external accessory may be mounted and used on a case of an electronic device to improve usability. To install the external accessory on the case, an adapter device for fixing the external accessory needs to be installed on the case.

[0221] Generally, the adapter device may be coupled to the case through an adhesive member such as a double-sided tape. However, when the adapter device is installed on the case through the adhesive member, there is a concern that the adhesive strength of the adhesive member may deteriorate with repeated use, causing the adapter device to be separated from the case.

[0222] According to various embodiments of the disclosure, an adapter accessory detachably coupled to a case of an electronic device may be provided.

[0223] However, the problems to be solved in the disclosure is not limited to the problem mentioned above, and may be determined in various ways without departing from the spirit and scope of the disclosure.

[0224] According to various embodiments of the disclosure, an adapter accessory may be configured to be easily coupled to or decoupled from a case of an electronic device.

[0225] The effects obtainable in the disclosure are not limited to the effects mentioned above, and other effects not mentioned may be clearly understood by those skilled in the art from the following description.

[0226] According to an example embodiment, the adapter accessory **100** may include the accessory body **110**, the deformable portion **120**, the button portion **130**, or the fixing portion **140**. The accessory body **110** may include the coupling portion **115** configured to couple the external accessory **20** thereto. The deformable portion **120** may extend from the accessory body. The deformable portion **120** may be configured to be at least partially deformable by a force applied from an outside. The button portion **130** may be connected to the deformable portion. The button portion **130** may be configured to press the deformable portion by the force applied from the outside or to be pressed by the deformable portion. The fixing portion **140** may be configured to be coupled, directly or indirectly, to the external case **10**. The fixing portion **140** may be formed to protrude from at least a portion of the button portion. A material of the deformable portion may be the same as a material of the button portion.

[0227] According to an embodiment, the accessory body may further include the first flat portion **111**, the first side portion **113**, or the first support wall **117**. The first flat portion **111** may have a circular shape. The first side portion **113** may extend from an edge of the first flat portion. The first support wall **117** may be formed to protrude from at least a portion of the first flat portion.

[0228] According to an embodiment, the first support wall may be disposed inside the first side portion.

[0229] According to an embodiment, the button portion may include the second flat portion **131** or the second side portion **133**. The second side portion **133** may extend from an edge of the second flat portion.

[0230] According to an embodiment, the pattern portion **134** including at least one slit may be formed on a surface of the second side portion.

[0231] According to an embodiment, the deformable portion may be formed to be at least partially bendable.

[0232] According to an embodiment, the deformable portion may include the first deformable part **121** or the second deformable part **123**. The first deformable part **121** may extend from a portion of the accessory body to a portion of the button portion. The second deformable part **123** may extend from another portion of the accessory body to another portion of the button portion.

[0233] According to an embodiment, the first deformable part may include the first inclined part **121**_a_ or the second inclined part **121**_b_. The second inclined part **121**_b_ may be obliquely connected to the first inclined part. The second deformable part may include the third inclined part **123***a* or the fourth inclined part **123***b*. The fourth inclined part **123***b* may be obliquely connected to the third inclined part.

[0234] According to an embodiment, a portion where the first inclined part **121***a* and the second inclined part **121***b* are connected may be disposed to face a portion where the third inclined part **123***a* and the fourth inclined part **123***b* are connected.

[0235] According to an embodiment, a portion where the first inclined part **221***a* and the second inclined part **221***b* are connected may be disposed to face an opposite direction to a portion where the third inclined part **223***a* and the fourth inclined part **223***b* are connected.

[0236] According to an embodiment, the deformable portion may include the first deformable part **321** or the first opening **322**. The first deformable part **321** may extend from a portion of the accessory body to a portion of the button portion. The first opening **322** may be formed to penetrate at least a portion of the first deformable part.

[0237] According to an embodiment, the deformable portion may include the first deformable part **421**, the second deformable part **423**, or the first opening **422**. The first deformable part **421** may extend from a portion of the accessory body to a portion of the button portion. The second deformable part **423** may extend from another portion of the accessory body to the portion of the button portion. The first opening **422** may be formed in at least a portion of the first deformable part.

[0238] According to an embodiment, the second deformable part may include the first inclined part **423***a* or the second inclined part **423***b*. The second inclined part may be obliquely connected to the first inclined part. At least a portion of the first inclined part may be accommodated in the first opening.

[0239] According to an embodiment, the deformable portion may include the first deformable part **521** or the second deformable part **523**. The first deformable part **521** may extend from a portion of the accessory body to a portion of the button portion. The second deformable part **523** may extend from another portion of the accessory body to the portion of the button portion. The first deformable part may include the first inclined part **521***a* or the second inclined part **521***b*. The first inclined part **521***a* may include the first opening **521***a***-1**. The second inclined part **521***b* may be obliquely connected to the first inclined part. The second inclined part **521***b* may include the second opening **521***b***-1**.

[0240] According to an embodiment, the second deformable part may be at least partially accommodated in the first opening or the second opening.

[0241] According to an embodiment, the fixing portion may include the inclined surface **140***a*. The inclined surface **140***a* may form an obtuse angle with respect to the button portion to which the fixing portion is connected.

[0242] According to an example embodiment, the adapter accessory **100** may include the accessory body **110**, the deformable portion **120**, the button portion **130**, or the fixing portion **140**. The accessory body **110** may be configured to be coupled to the external accessory **20**. The deformable portion **120** may extend from the accessory body. The deformable portion **120** may be configured to be at least partially deformable by a force applied from an outside. The button portion **130** may be connected to the deformable portion. The button portion **130** may press the deformable portion by the force applied from the outside or be pressed by the deformable portion. The fixing portion **140** may be configured to be coupled to an external case. The fixing portion **140** may be formed to protrude from at least a portion of the button portion. The accessory body, the deformable portion, the button portion, and the fixing portion may be formed integrally.

[0243] According to an embodiment, the accessory body may further include the first flat portion **111**, the first side portion **113**, or the first support wall **117**. The first flat part **111** may have a circular shape. The first side portion **113** may extend from an edge of the first flat portion. The first support wall **117** may be formed to protrude from at least a portion of the first flat portion.

[0244] According to an embodiment, the button portion may include the second flat portion **131** or the second side portion **133**. The second side portion **133** may extend from an edge of the second flat portion.
[0245] According to an embodiment, the pattern portion **134** including at least one slit may be formed on a surface of the second side portion.
[0246] While specific embodiments have been described in the detailed description of the disclosure, it will be apparent to those skilled in the art that many modifications may be made within the scope of the disclosure.
[0247] While the disclosure has been illustrated and described with reference to various embodiments, it will be understood that the various embodiments are intended to be illustrative, not limiting. It will further be understood by those skilled in the art that various changes in form and detail may be made without departing from the true spirit and full scope of the disclosure, including the appended claims and their equivalents. It will also be understood that any of the embodiment(s) described herein may be used in conjunction with any other embodiment(s) described herein.

Claims

1. An adapter accessory comprising: an accessory body including a coupling portion configured to couple an external accessory thereto; a deformable portion, comprising flexible material, extending from the accessory body and configured to be at least partially deformable by a force applied from an outside; a button portion connected to the deformable portion and configured to press the deformable portion due to the force applied from the outside and/or to be pressed by the deformable portion; and a fixing portion, comprising a protrusion, configured to be coupled to an external case and formed to protrude from at least part of the button portion, wherein a material of the deformable portion is the same as a material of the button portion.
2. The adapter accessory of claim 1, wherein the accessory body further includes: a first flat portion having a circular shape; a first side portion extending from an edge of the first flat portion; and a first support wall formed to protrude from at least of the first flat portion.
3. The adapter accessory of claim 2, wherein the first support wall is disposed inside the first side portion.
4. The adapter accessory of claim 1, wherein the button portion includes: a second flat portion; and a second side portion extending from an edge of the second flat portion.
5. The adapter accessory of claim 4, wherein a pattern portion including at least one slit is formed on a surface of the second side portion.
6. The adapter accessory of claim 1, wherein the deformable portion is configured to be at least partially bendable.
7. The adapter accessory of claim 1, wherein the deformable portion includes: a first deformable part extending from a portion of the accessory body to a portion of the button portion; and a second deformable part extending from another portion of the accessory body to another portion of the button portion.
8. The adapter accessory of claim 7, wherein the first deformable part includes a first inclined part and a second inclined part obliquely connected to the first inclined part, and wherein the second deformable part includes a third inclined part and a fourth inclined part obliquely connected to the third inclined part.
9. The adapter accessory of claim 8, wherein a portion where the first inclined part and the second inclined part are connected is disposed to face a portion where the third inclined part and the fourth inclined part are connected, respectively.
10. The adapter accessory of claim 8, wherein a portion where the first inclined part and the second inclined part are connected is disposed to face an opposite direction to a portion where the third inclined part and the fourth inclined part are connected, respectively.

11. The adapter accessory of claim 1, wherein the deformable portion includes: a first deformable part extending from a portion of the accessory body to a portion of the button portion; and a first opening formed to penetrate at least a portion of the first deformable part.
12. The adapter accessory of claim 1, wherein the deformable portion includes: a first deformable part extending from a portion of the accessory body to a portion of the button portion; a second deformable part extending from another portion of the accessory body to the portion of the button portion; and a first opening formed in at least a portion of the first deformable part.
13. The adapter accessory of claim 12, wherein the second deformable part includes a first inclined part and a second inclined part obliquely connected to the first inclined part, and wherein at least a portion of the first inclined part is accommodated in the first opening.
14. The adapter accessory of claim 1, wherein the deformable portion includes: a first deformable part extending from a portion of the accessory body to a portion of the button portion; and a second deformable part extending from another portion of the accessory body to the portion of the button portion, and wherein the first deformable part includes a first inclined part including a first opening and a second inclined part obliquely connected to the first inclined part and including a second opening.
15. The adapter accessory of claim 14, wherein the second deformable part is at least partially accommodated in the first opening or the second opening.
16. The adapter accessory of claim 1, wherein the fixing portion includes an inclined surface forming an obtuse angel with the button portion connected to the fixing member.
17. An adapter accessory comprising: an accessory body including a coupling portion configured to couple an external accessory thereto; a deformable portion, comprising flexible material, extending from the accessory body and configured to be at least partially deformable by a force applied from an outside; a button portion connected to the deformable portion and configured to press the deformable portion due to the force applied from the outside and/or to be pressed by the deformable portion; and a fixing portion, comprising a protrusion, configured to be coupled to an external case and formed to protrude from at least of the button portion, wherein the accessory body, the deformable portion, the button portion and the fixing portion are formed as a single piece.
18. The adapter accessory of claim 17, wherein the accessory body further comprises; a first flat portion having a circular shape; a first side portion extending from an edge of the first flat portion; and a first support wall formed to protrude from at least of the first flat portion.
19. The adapter accessory of claim 17, wherein the button portion includes: a second flat portion; and a second side portion extending from an edge of the second flat portion.
20. The adapter accessory of claim 19, wherein a pattern portion including at least one slit is formed on a surface of the second side portion.